# Ethanol Blending in Gasoline – Portugal

September 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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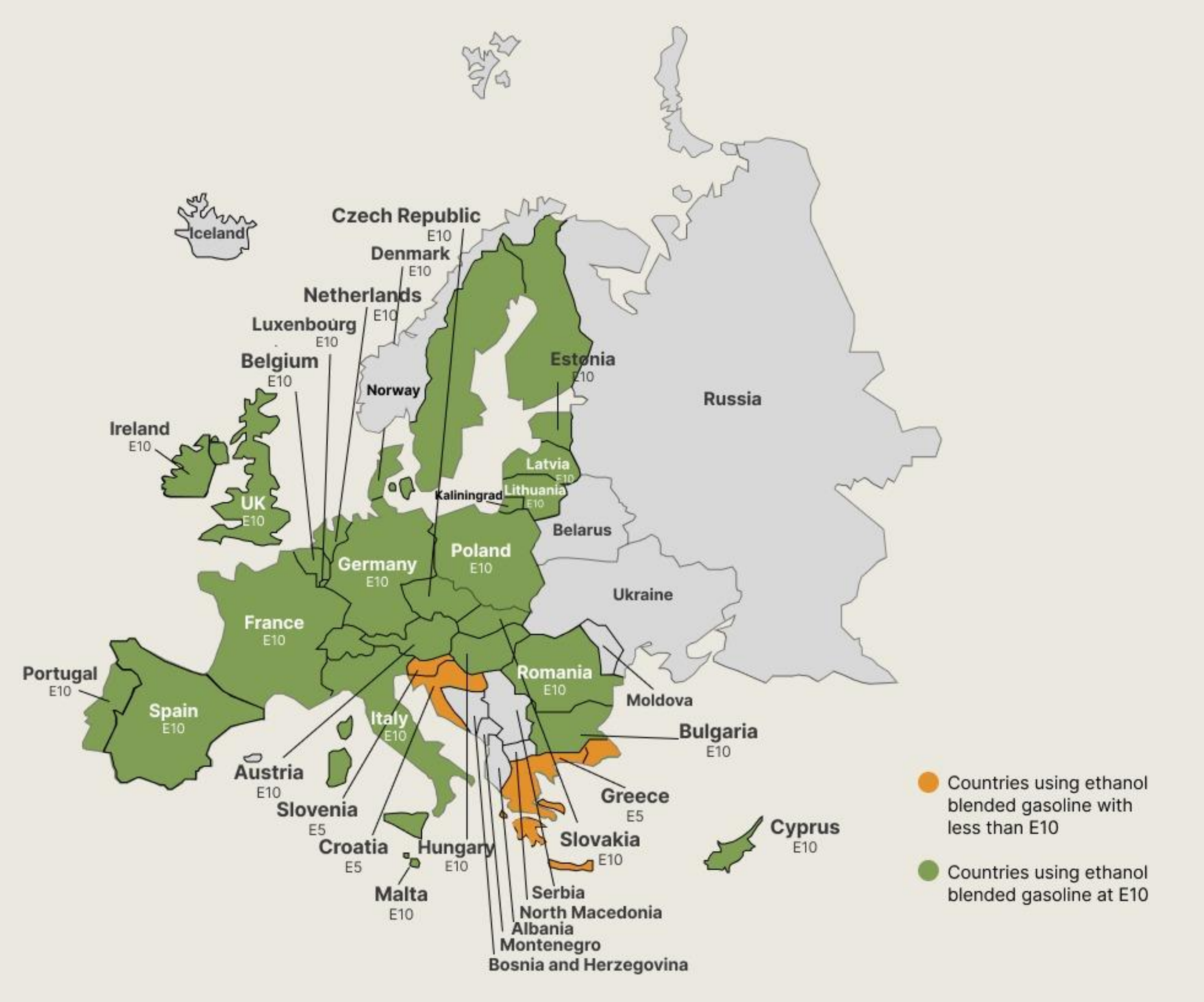
# Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.




Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

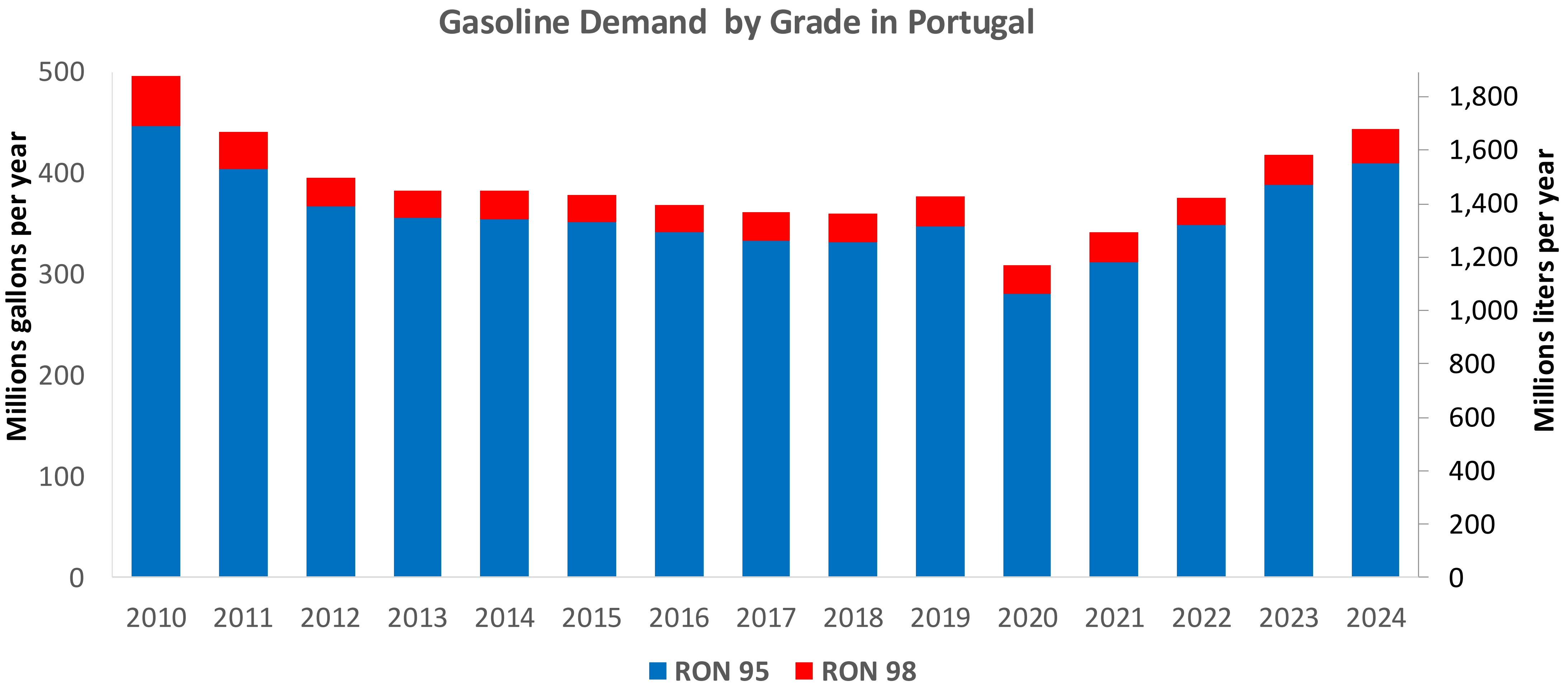
# Ethanol Gasoline Blending in Portugal


In 2024, gasoline consumption in Portugal was 1,688 million liters (446 million gallons) and growth rate was 0.2%. Gasoline production and net imports were 2,990 and -1,310 million liters (790 and -346 million gallons) correspondingly. Portugal complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Portugal was 64 million liters (17 million gallons) representing 3.8% of gasoline demand. Portugal does not produce ethanol, supply is done through imports. Local production plants are being built.

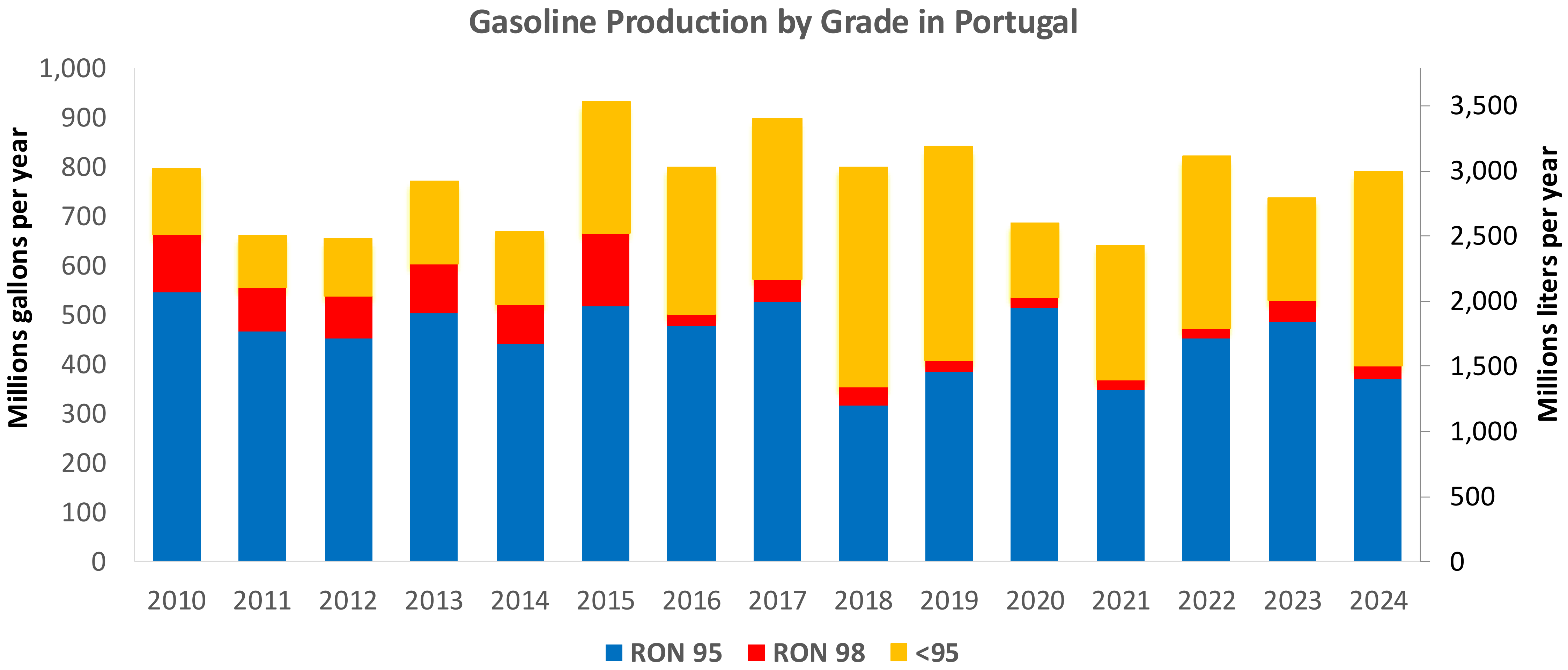
Source: Eurostat, European Commision

# Gasoline Demand in Portugal



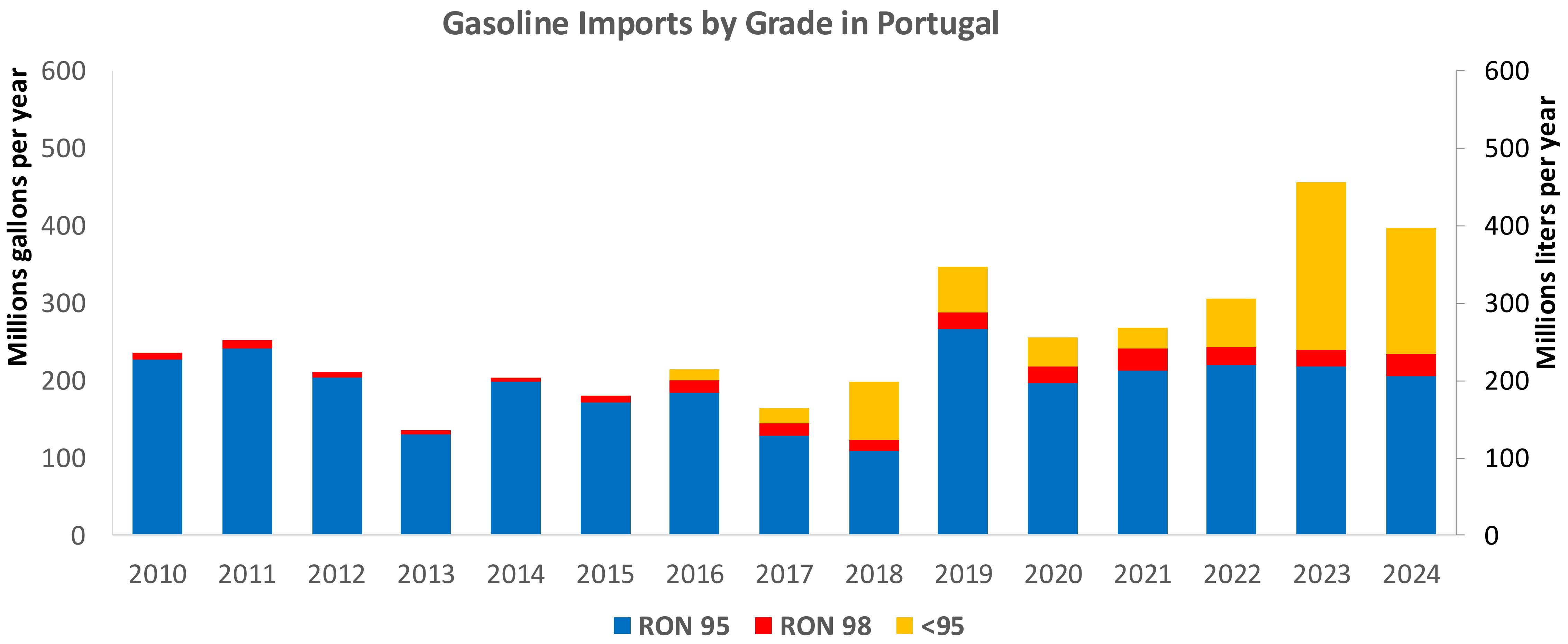

Source: Eurostat

# Gasoline Production in Portugal



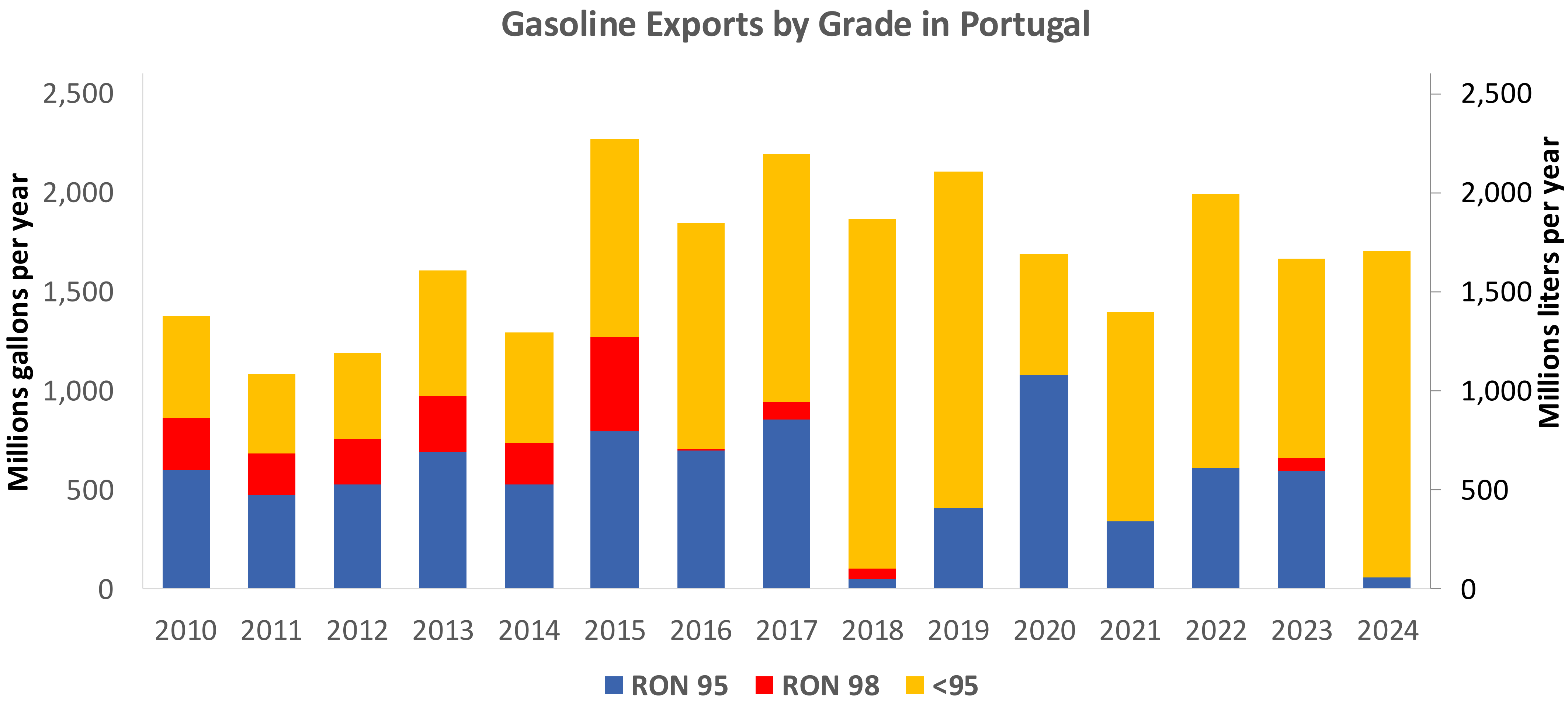

Source: Eurostat

# Gasoline Imports in Portugal



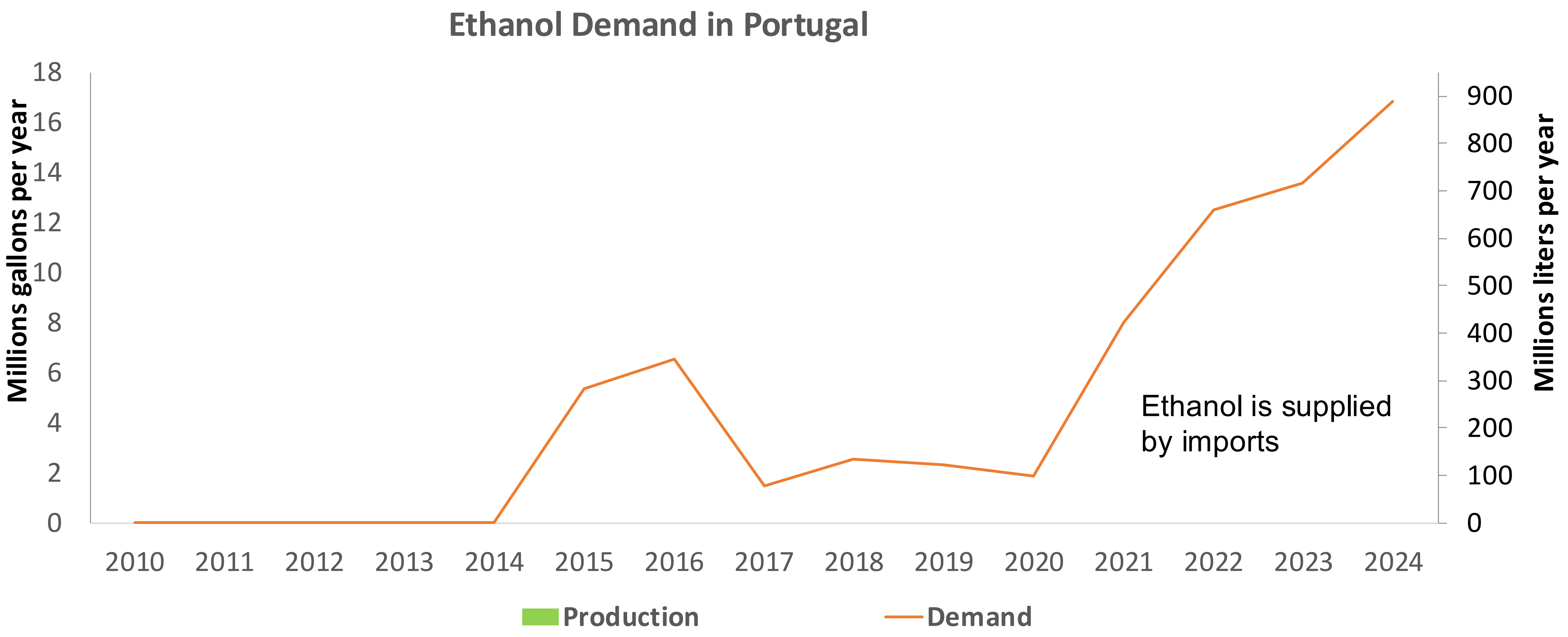

Source: Eurostat

# Gasoline Exports in Portugal




Source: Eurostat

# Ethanol Balance in Portugal




Source: Eurostat

# Gasoline Quality in Portugal

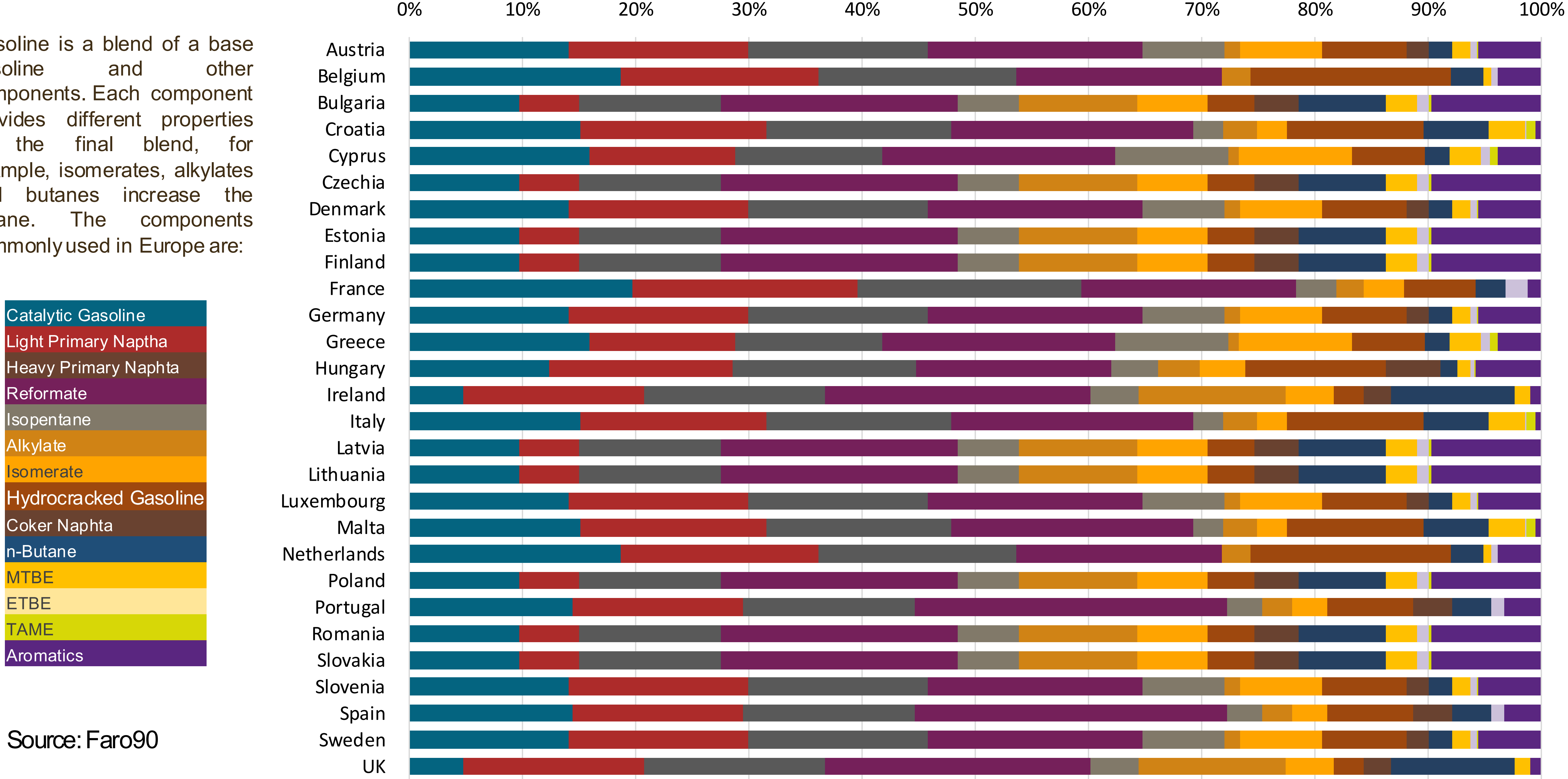
| Name | EN 228:2025 (Euro 6); Country Level Fuel Ordinances | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Implementation Date | 2025 | | | | | | | |
| Applicability | All EU Contries + UK | | | | | | | |
| Grade | RON 95 E5 | | RON 95 E10 | | RON 98 E5 | | RON 98 E10 | |
| | Min | Max | Min | Max | Min | Max | Min | Max |
| RON | 95.0 | | 95.0 | | 98.0 | | 98.0 | |
| MON | > 85 | > 88 | > 85 | > 88 | > 85 | > 88 | > 85 | > 88 |
| AKI | | | | | | | | |
| Benzene (%vol) | < 1 | | | | | | | |
| Aromatics (% vol) | < 35 | | | | | | | |
| Olefins (%vol) | < 18 | | | | | | | |
| Lead (g/L) | < 0.005 | | | | | | | |
| Manganese (mg/L) | < 2,0 | | | | | | | |
| Sulfur (ppm) | 10.0 | | | | | | | |
| Oxygen (%w)* | < 2,7 | < 3,7 | < 2,7 | < 3,7 | < 2,7 | < 3,7 | < 2,7 | < 3,7 |
| Ethanol (%vol) | < 5 | < 10 | < 5 | < 10 | < 5 | < 10 | < 5 | < 10 |
| PVR 37.8°C (psi) | <> 45 - 100 kPa in 6 Volatility Classes | | | | | | | |
| Overall Biofuels (%cal) | 13.0 | | | | | | | |
| Advanced Biofuels (%cal) | 2.0 | | | | | | | |
| Bioethanol in Gasoline Sales (%cal) | - | | | | | | | |
| GHG Emission Reduction (%) | - | | | | | | | |

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

# Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:




Source: Faro90

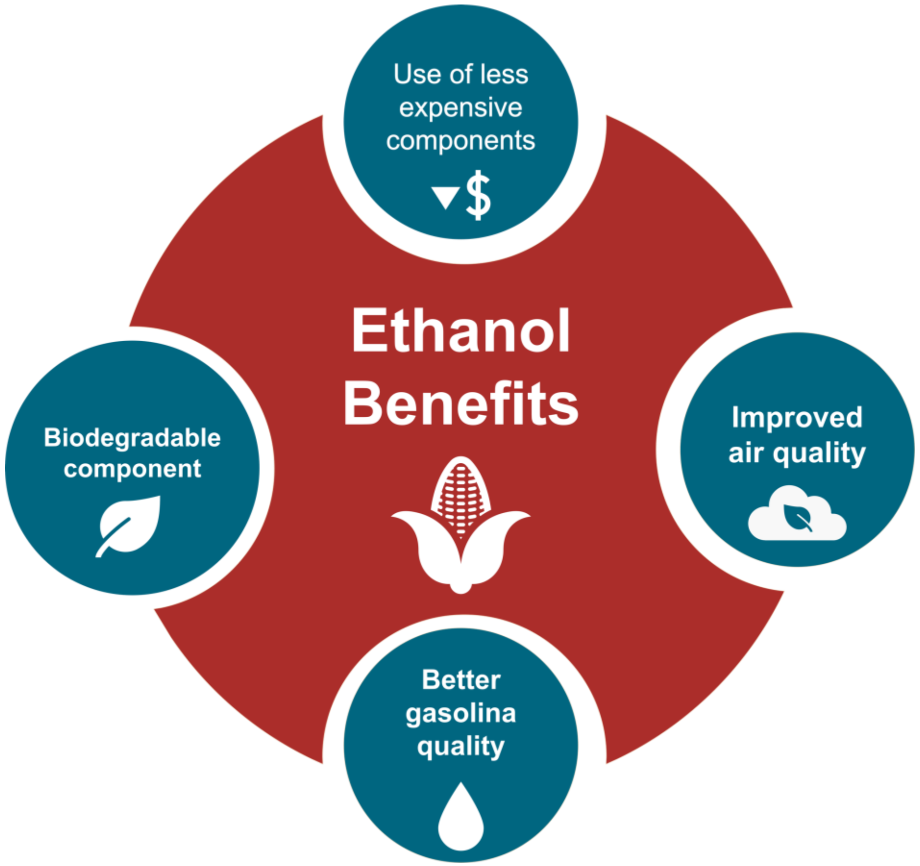
# Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

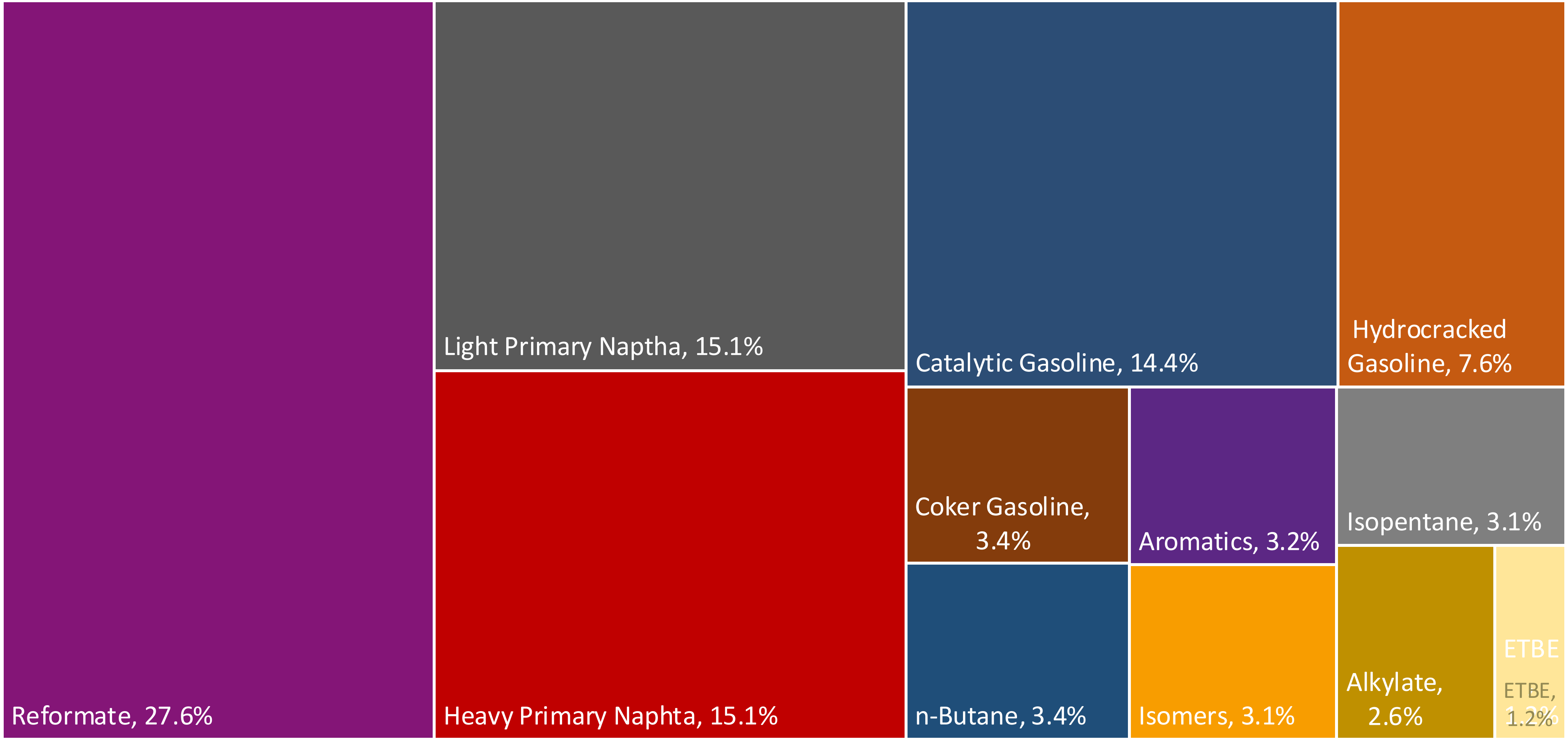
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

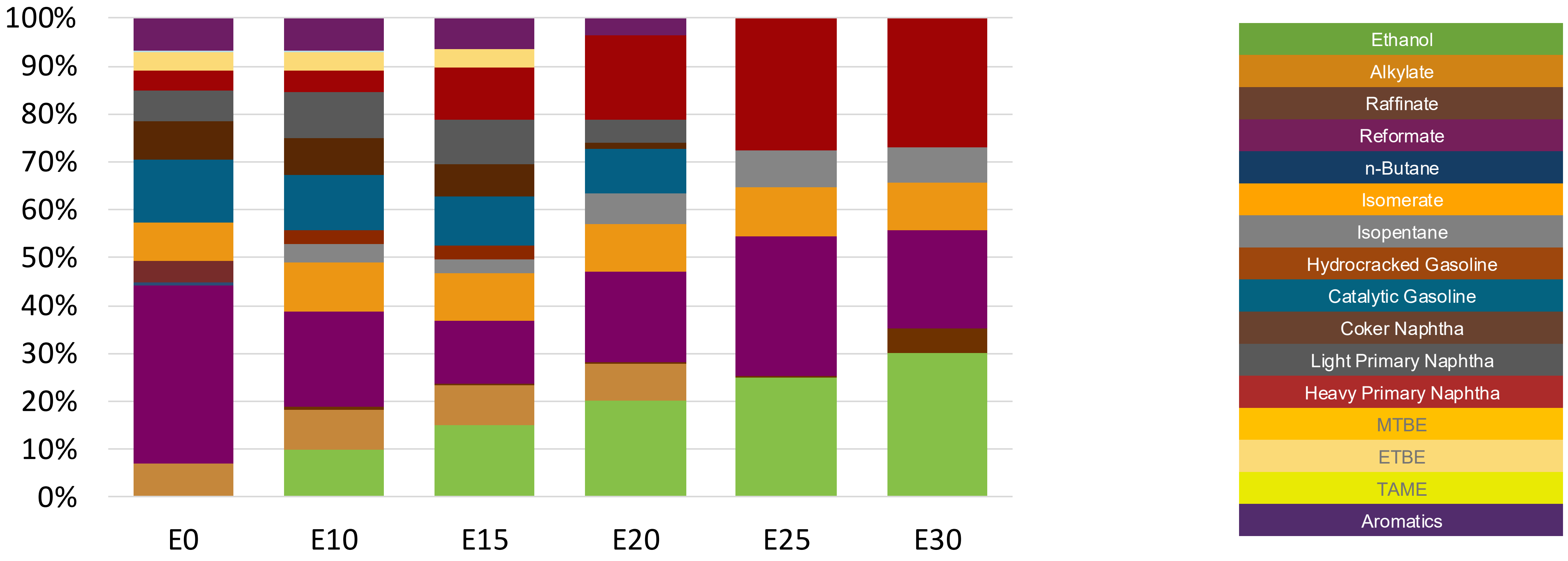


# Portugal Available Blending Components




Source: Faro90

# Portugal – Gasoline 95 – Constant Octane

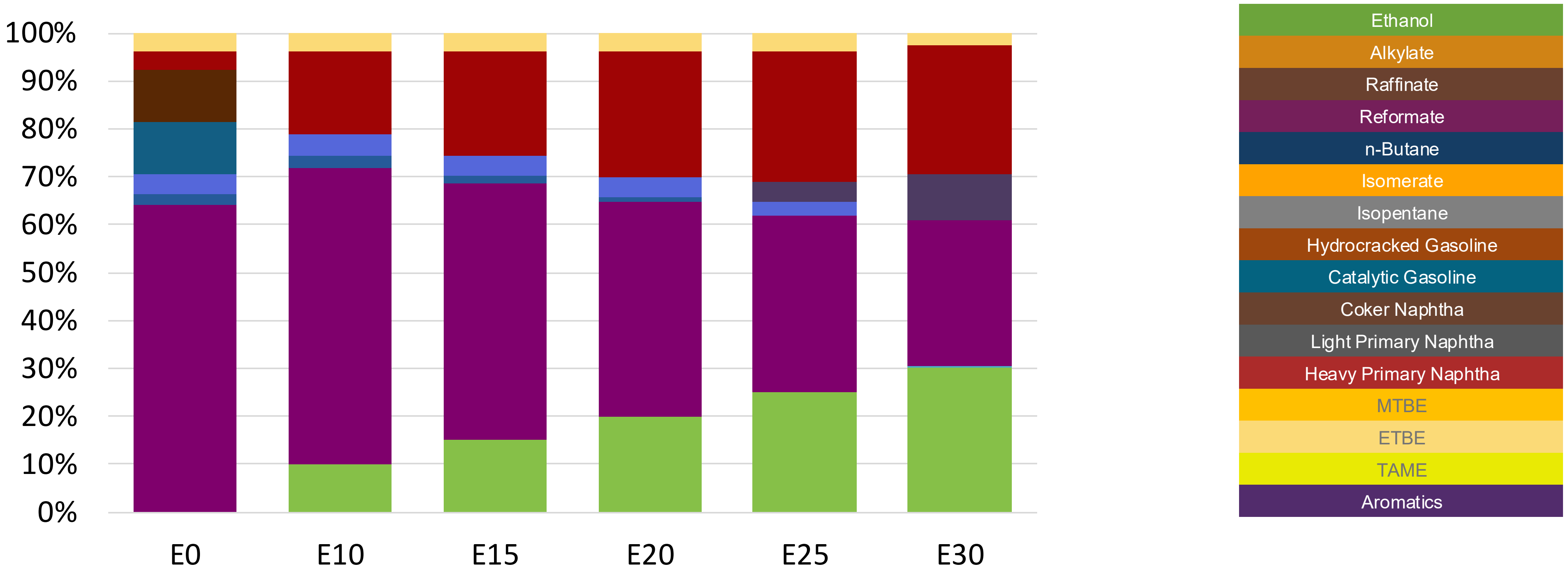



| | | | | | | |
|---|---|---|---|---|---|---|
| Octane (RON) | 95.0 | 95.0 | 95.0 | 95.0 | 95.3 | 95.8 |
| Price (US$/gal) | $ 2.290 | $ 2.178 | $ 2.098 | $ 2.008 | $ 1.893 | $ 1.830 |

Average CIF 2024 Prices

Source: Faro90

# Portugal - Gasoline 98 - Constant Octane

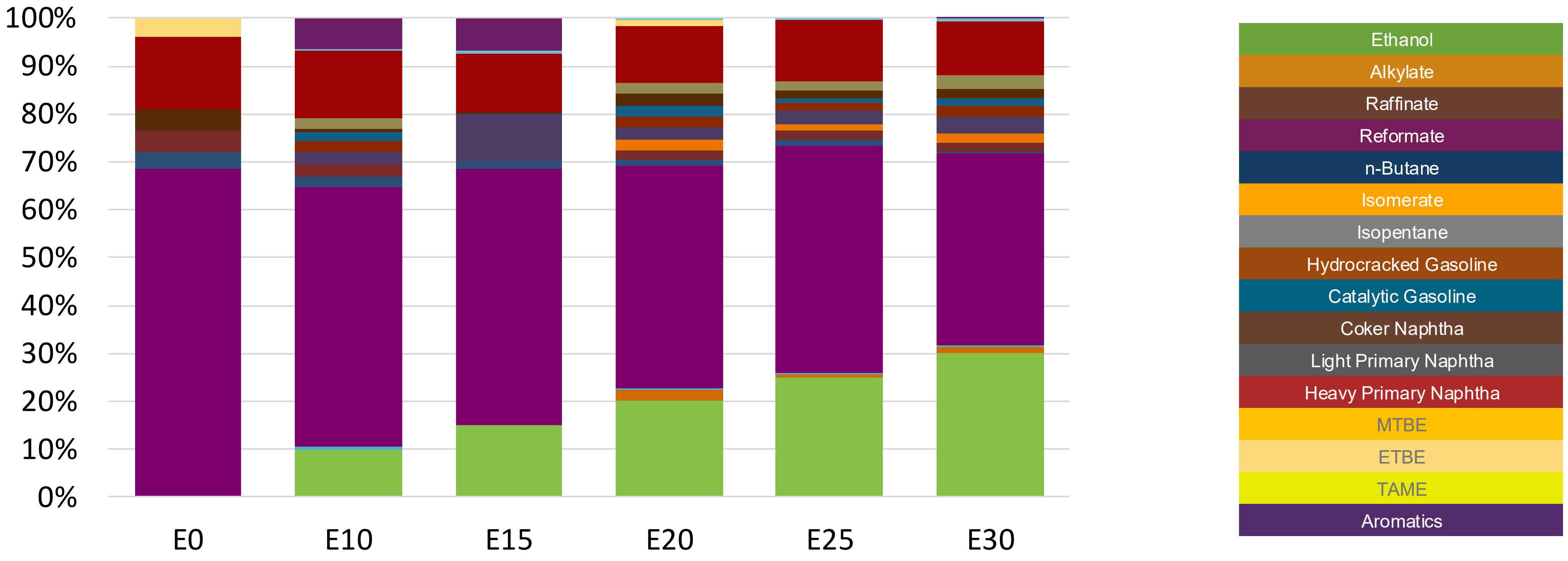



| | | | | | | |
|---|---|---|---|---|---|---|
| Octane (RON) | 98.0 | 98.0 | 98.0 | 98.0 | 98.0 | 98.0 |
| Price (US$/gal) | $ 2.359 | $ 2.235 | $ 2.166 | $ 2.091 | $ 2.013 | $ 1.934 |

Average CIF 2024 Prices

Source: Faro90

# Portugal – Gasoline 95 – Increased Octane

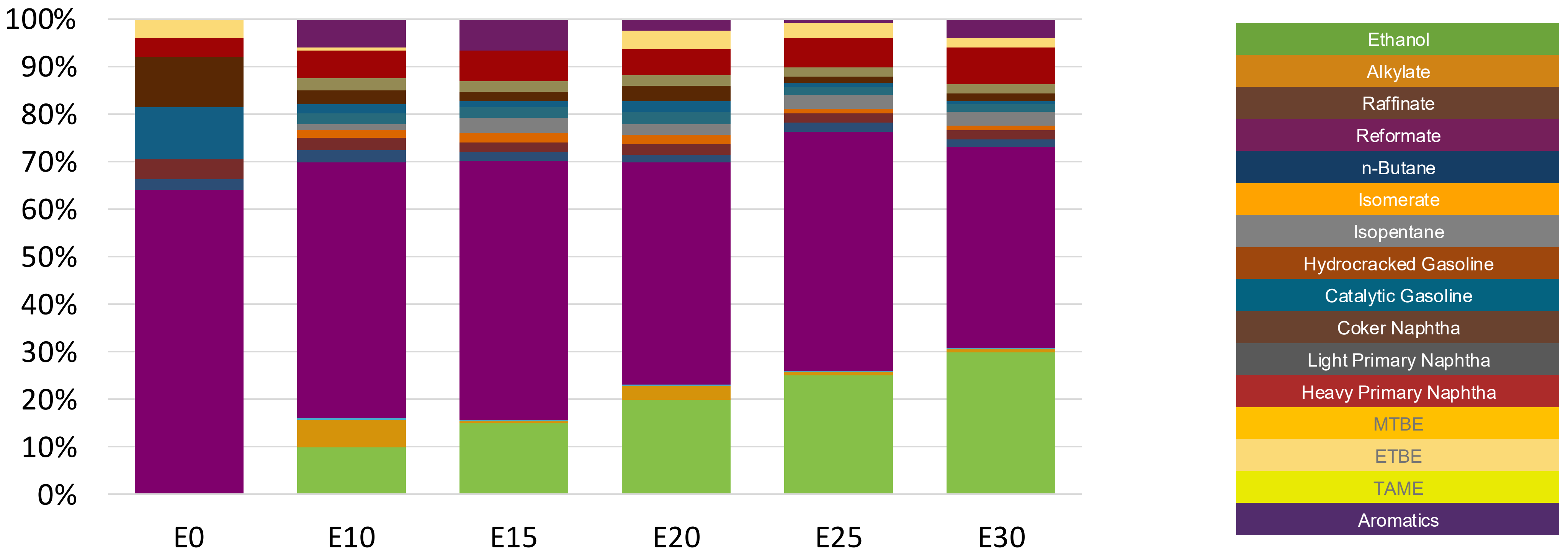



| | | | | | | |
|---|---|---|---|---|---|---|
| Octane (RON) | 95.0 | 96.8 | 98.2 | 99.9 | 101.4 | 102.6 |
| Price (US$/gal) | $ 2.222 | $ 2.222 | $ 2.222 | $ 2.222 | $ 2.222 | $ 2.222 |

Average CIF 2024 Prices

Source: Faro90

# Portugal – Gasoline 98 – Increased Octane




| | | | | | | |
|---|---|---|---|---|---|---|
| Octane (RON) | 98.0 | 99.3 | 100.9 | 102.5 | 104.0 | 105.0 |
| Price (US$/gal) | $ 2.359 | $ 2.359 | $ 2.359 | $ 2.359 | $ 2.359 | $ 2.359 |

Average CIF 2024 Prices

Source: Faro90

# Vehicle Emission Impact for Ethanol Gasoline Blending

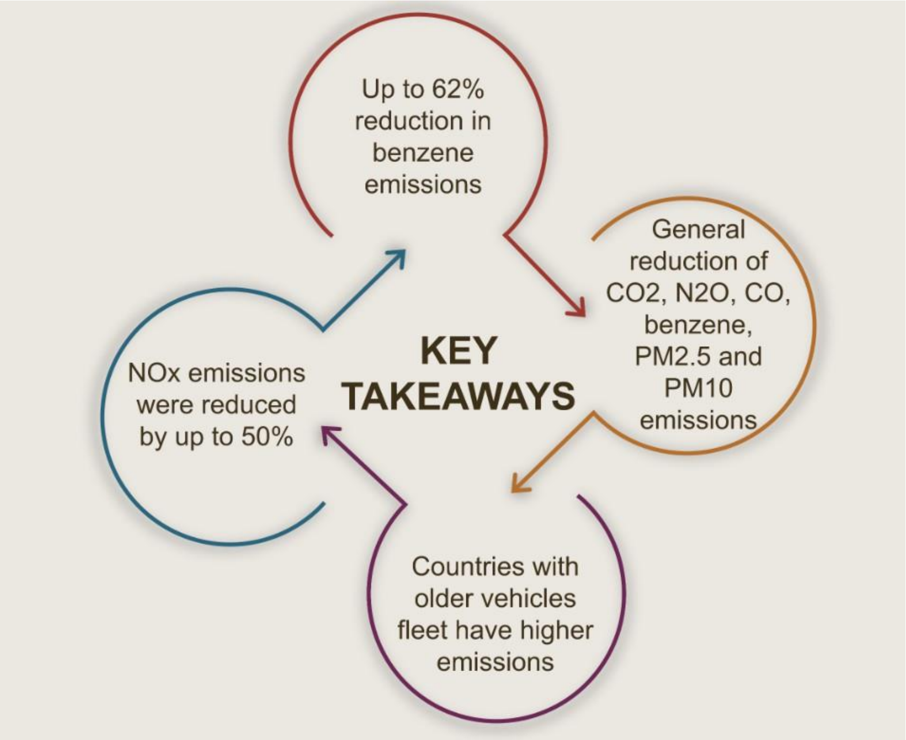
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results




**Source: Bureau of transportation statistics.*

# Gasoline Vehicle Fleet - Portugal

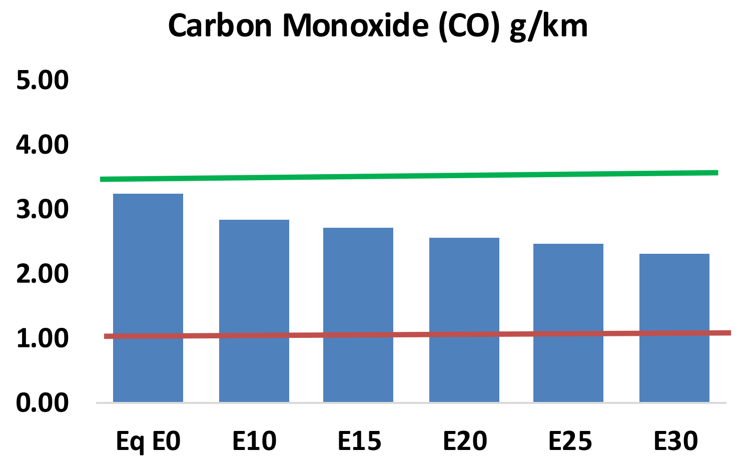
| Type | Weight | Motor | Escape | Recover | Age | No. Vehicles |
|---|---|---|---|---|---|---|
| Auto/Sml Truck | Light | Carburator | 3-Ways | PCV Tank | >8 years | 210 |
| | | Single FI | 3-Ways | | >8 years | 48,136 |
| | | Multi FI | 3-Ways | | >8 years | 610,603 |
| | | | Euro III | | >8 years | 787,083 |
| | | | Euro IV | | >8 years | 1,089,910 |
| | | | Euro V | | >8 years | 241,125 |
| | | | Euro 6 | | >8 years | 626,653 |
| | | | | | 4-8 years | 815,902 |
| | | | | | <4 years | 706,256 |
| | | | Hybrid | | All | 47,501 |
| | | Electric | | | All | 37,123 |
| | Medium | Carburator | 3-Ways | PCV Tank | >8 years | 212 |
| | | Single FI | 3-Ways | | >8 years | 34,282 |
| | | Multi FI | 3-Ways | | >8 years | 207,932 |
| | | | Euro III | | >8 years | 296,136 |
| | | | Euro IV | | >8 years | 306,262 |
| | | | Euro V | | >8 years | 49,391 |
| | | | Euro 6 | | >8 years | 108,733 |
| | | | | | 4-8 years | 143,188 |
| | | | | | <4 years | 111,843 |
| | | | Hybrid | | All | 5 |
| | | Electric | | | All | 2,303 |
| Sml Engines | Light | 4-Cycle | None | PCV | >8 years | 18,953 |
| | | | Euro III | | >8 years | 102,702 |
| | | | Euro IV | | >8 years | 657,528 |
| | | | Euro V | | >8 years | 35,767 |
| | | | Euro 6 | | >8 years | 67,293 |
| | | | | | 4-8 years | 120,172 |
| | | | | | <4 years | 162,209 |
| | | | Hybrid | | All | 0 |
| | | Electric | | | All | 13,219 |
| Truck | Heavy | Carburator | 3-Ways | PCV | >8 years | 34 |
| | | Fuel-Injection | 3-Ways | | >8 years | 21,062 |
| | | | Euro III | | >8 years | 17,672 |
| | | | Euro IV | | >8 years | 24,223 |
| | | | Euro V | | >8 years | 4,467 |
| | | | Euro 6 | | >8 years | 14,232 |
| | | | | | 4-8 years | 22,182 |
| | | | | | <4 years | 25,555 |
| | | | Hybrid | | All | 0 |
| | | Electric | | | All | 153 |

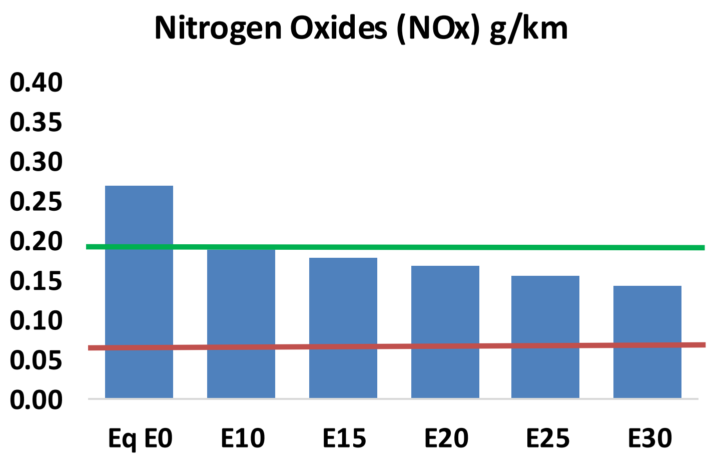
Vehicle Fleet: 7,578,214
Gasoline Vehicle Fleet: 2,654,720

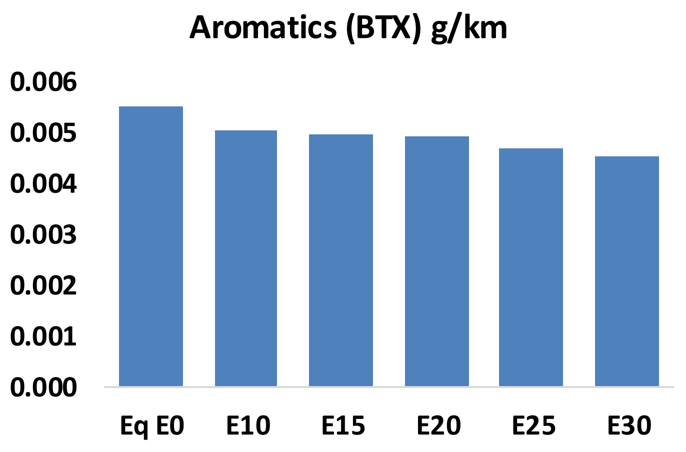
Source: Eurostat, ACEA

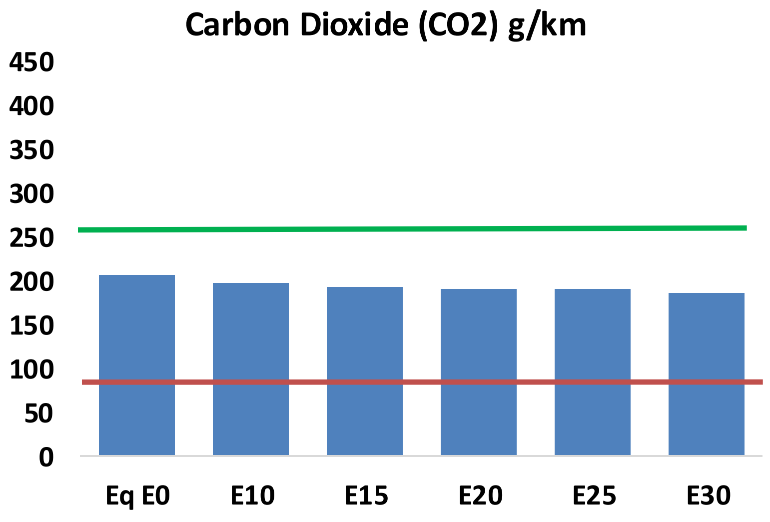
# Portugal – Vehicular Emissions

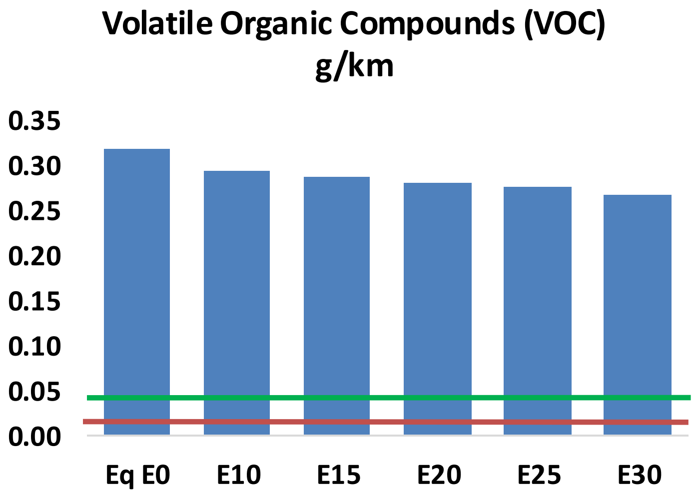
TIER III BIN 125 United States
Euro 6 Standard

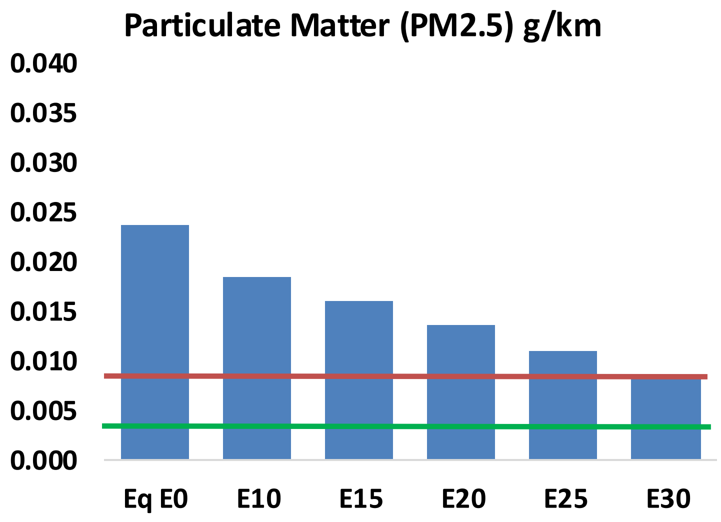



Source: Faro90

# Portugal – Vehicular Emissions

| Vehicular Fleet Emissions (kg/day) | Eq E0 | E5 | E10 | E15 | E20 | E25 | E30 | Reduction (%) | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | E10 | E20 |
| **CO** | 209,074 | 196,659 | 184,244 | 174,785 | 165,594 | 158,626 | 150,367 | -12% | -21% |
| **CO2** | 13,460,947 | 13,114,808 | 12,787,900 | 12,530,796 | 12,403,570 | 12,331,260 | 12,103,959 | -5% | -8% |
| **VOC** | 20,658 | 19,868 | 19,077 | 18,586 | 18,162 | 17,858 | 17,377 | -8% | -12% |
| **NOx** | 17,465 | 13,098 | 12,225 | 11,522 | 10,866 | 10,143 | 9,379 | -30% | -38% |
| **SOx** | 160 | 148 | 136 | 125 | 116 | 105 | 94 | -15% | -28% |
| **NH3** | 4,599 | 4,553 | 4,508 | 4,529 | 4,580 | 4,616 | 4,646 | -2% | 0% |
| **N2O** | 175 | 174 | 173 | 174 | 178 | 180 | 182 | -1% | 2% |
| **CH4** | 4,900 | 4,900 | 4,900 | 4,974 | 5,082 | 5,175 | 5,263 | 0% | 4% |
| **Lead** | - | - | - | - | - | - | - | 0% | 0% |
| **Butadiene** | 198 | 193 | 187 | 184 | 182 | 181 | 178 | -6% | -8% |
| **Acetaldehyde** | 421 | 483 | 708 | 1,079 | 1,470 | 1,679 | 1,984 | 68% | 249% |
| **Formaldehyde** | 1,337 | 1,300 | 1,516 | 1,780 | 1,864 | 2,062 | 2,245 | 13% | 39% |
| **Benzene** | 358 | 344 | 326 | 321 | 319 | 305 | 295 | -9% | -11% |
| **PM2.5** | 839 | 745 | 651 | 566 | 480 | 390 | 295 | -22% | -43% |
| **PM10** | 690 | 613 | 536 | 465 | 395 | 320 | 243 | -22% | -43% |

Source: Faro90

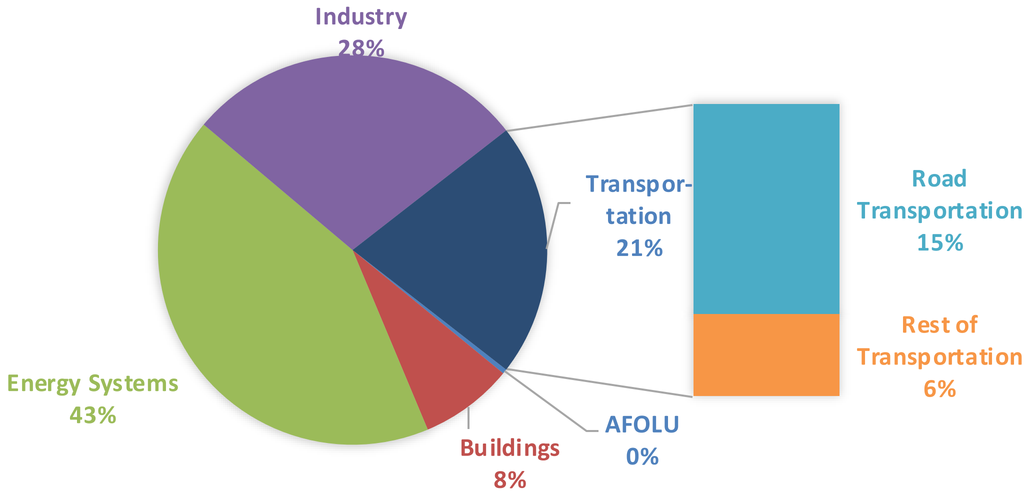
# Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

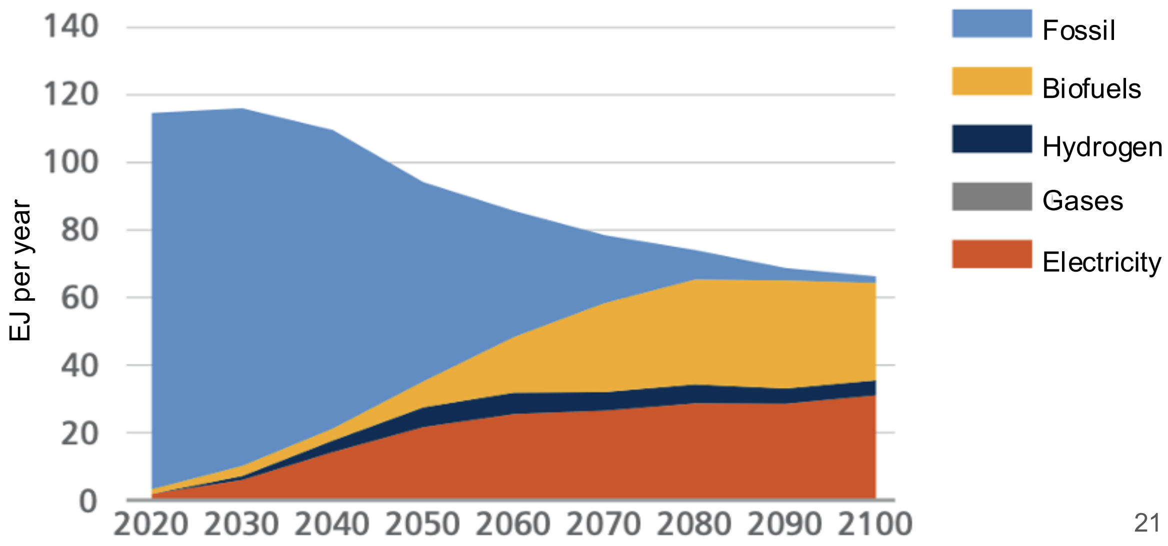
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

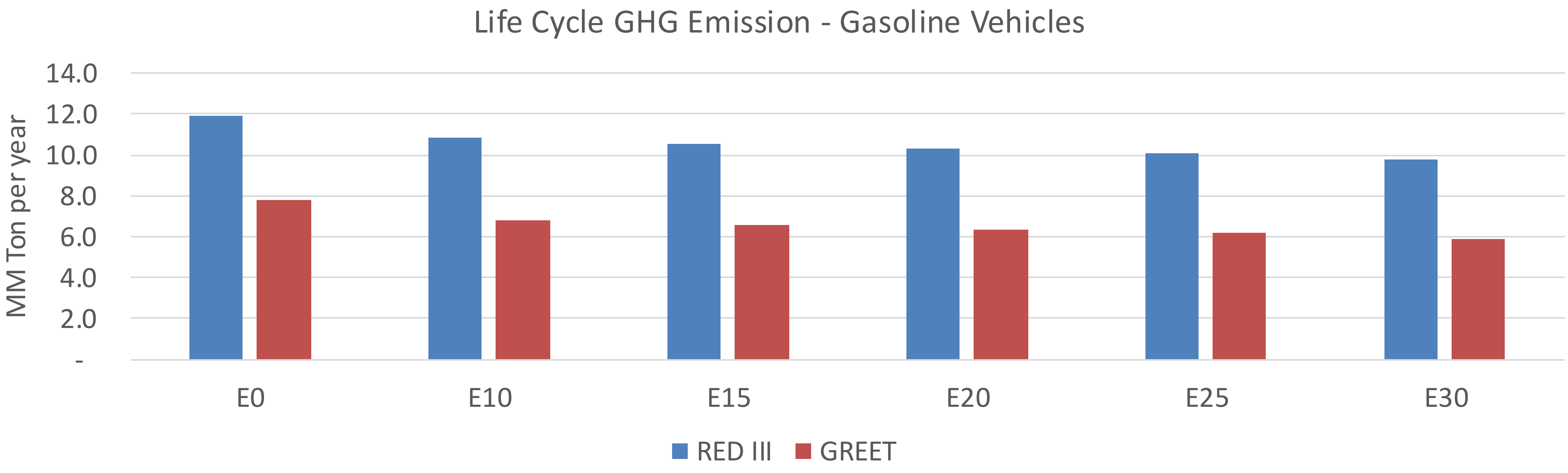
**Current GHG Global Emissions Distribution**




**IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023**



# Portugal – Gasoline Vehicle Fleet Life Cycle GHG Emissions




| | |
|---|---|
| Country Emissions 2024 (MMT/año) | 59.0 |
| Target @2035 (MMT/year) | 33.2 |
| Delta | 25.8 |

| %Reduction vs E0 | E0 | E10 | E15 | E20 | E25 | E30 |
|---|---|---|---|---|---|---|
| GREET 2024 | 0.0% | 12.6% | 15.8% | 18.5% | 21.0% | 24.4% |
| RED II/III | 0.0% | 9.1% | 11.6% | 13.7% | 15.7% | 18.2% |

| % Target@2035 | E0 | E10 | E15 | E20 | E25 | E30 |
|---|---|---|---|---|---|---|
| GREET 2024 | 0.0% | 3.8% | 4.8% | 5.6% | 6.4% | 7.4% |
| RED II/III | 0.0% | 4.2% | 5.4% | 6.3% | 7.3% | 8.4% |

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90